REAL-TIME INTRUSION DETECTION SYSTEM

Machine Learning Network Security Report

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1. Executive Summary

This report summarizes the network traffic analyzed by the Real-Time Intrusion Detection System. The system uses machine-learning-based classification to identify network flows as either normal traffic or potential intrusions.

Metric	Value
Total Network Flows	6802
Normal Flows	6340
Detected Attacks	462
Attack Rate	6.79%

2. Machine Learning Model

Parameter	Value
Algorithm	Random Forest Classifier
Training Features	duration, protocol_type, src_bytes, dst_bytes, count
Classification	Normal / Attack
Optimized Threshold	0.30
Validation Accuracy	80.96%
Attack Precision	0.93
Attack Recall	0.72
Attack F1-Score	0.81

3. Network Protocol Distribution

Protocol	Flow Count
TCP	3738
UDP	3062
OTHER	2

4. Detected Attack History

Time	Source	Destination	Protocol	Confidence
2026-08-25 19:02:11	10.2.0.1:65432	192.168.1.18:64028	TCP	42.68%
2026-08-25 19:01:57	150.171.110.85:443	192.168.1.18:50082	TCP	53.53%
2026-08-25 19:01:33	192.168.1.18:53035	202.88.211.152:443	UDP	30.41%
2026-08-25 19:01:11	192.168.1.18:53261	192.178.211.84:443	UDP	31.42%
2026-08-25 19:01:06	10.2.0.1:65432	192.168.1.18:61881	TCP	42.68%
2026-08-25 19:00:24	192.168.1.18:52900	192.178.211.84:443	UDP	30.12%
2026-08-25 18:59:57	10.2.0.1:65432	192.168.1.18:65038	TCP	42.48%
2026-08-25 18:59:44	192.168.1.18:64929	4.2.2.2:53	UDP	74.82%
2026-08-25 18:59:40	192.168.1.18:52611	8.8.8.8:53	UDP	74.82%
2026-08-25 18:58:47	10.2.0.1:65432	192.168.1.18:52630	TCP	42.68%
2026-08-25 18:58:01	172.64.155.209:443	192.168.1.18:60490	TCP	31.03%
2026-08-25 18:57:43	10.2.0.1:65432	192.168.1.18:56164	TCP	42.68%
2026-08-25 18:57:23	142.251.10.132:443	192.168.1.18:59014	UDP	38.66%
2026-08-25 18:57:21	142.251.155.119:443	192.168.1.18:59446	UDP	81.74%
2026-08-25 18:57:07	142.250.183.142:443	192.168.1.18:60000	UDP	33.3%
2026-08-25 18:57:06	172.253.118.113:443	192.168.1.18:63565	UDP	31.39%
2026-08-25 18:57:03	142.250.66.14:443	192.168.1.18:61970	UDP	53.7%
2026-08-25 18:57:02	142.251.223.14:443	192.168.1.18:55345	UDP	39.71%
2026-08-25 18:56:57	104.18.39.21:443	192.168.1.18:51830	TCP	48.13%
2026-08-25 18:56:55	150.171.110.85:443	192.168.1.18:51990	TCP	54.27%

5. Security Recommendations

1. Investigate repeated attack predictions from the same source address.

2. Monitor network flows with unusually high traffic volume or packet counts.

3. Review suspicious TCP and UDP connections identified by the ML classifier.

4. Use the attack confidence score together with network context when validating alerts.

5. Continue collecting representative network traffic to improve the IDS model.

6. Periodically retrain and evaluate the model using updated datasets.

6. Conclusion

The Real-Time Intrusion Detection System combines live network packet capture, flow-level feature extraction, machine-learning classification, threshold optimization, CSV logging and dashboard-based monitoring. The generated statistics and attack history provide a summary of the traffic observed during the monitoring period.